Geography Ch-13 (Must-Do): Weather - Insolation, Pressure, Winds, Humidity & Rainfall

Tier 1: Must-Do | Source: GC Leong (Certificate Physical & Human Geography) + Majid Husain (Indian & World Geography)

Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

Elements of Weather: Insolation, Pressure Belts, Planetary Winds, Humidity & Rainfall

GS-I (Physical Geography - Climatology)
Geography

■ NEWS TRIGGER

Foundation concept (GC Leong Ch-11/12; Majid Husain Climatology). Weather = short-term state of the atmosphere at a place; described by 5 elements - temperature (insolation), pressure, wind, humidity and precipitation.

■ INTRO & ORIGIN

Insolation (incoming solar radiation) is the ultimate driver of weather. Uneven heating of the earth (latitude, tilt, day length) creates temperature differences, which create pressure differences, which drive winds; winds carry moisture (humidity) which condenses to give rainfall. This chain is the core of climatology.

Heat budget: the earth as a whole maintains balance - incoming shortwave solar radiation is balanced by outgoing terrestrial longwave radiation. Surplus in tropics (0-40 deg) and deficit in higher latitudes (40-90 deg) is evened out by winds and ocean currents (heat transfer poleward).

■ KEY DATA

Element	Definition / Driver	Key Facts
Insolation	Incoming solar radiation received at surface	Max at equator (vertical sun); decreases poleward; modified by cloud, aspect, altitude
Pressure belts	Weight of air column (thermal + dynamic)	Equatorial Low, Sub-tropical High (30 deg), Sub-polar Low (60 deg), Polar High
Planetary winds	Blow from High to Low pressure, deflected by Coriolis	Trades (E), Westerlies (W), Polar easterlies
Humidity	Water vapour in air (absolute / relative)	Relative humidity = actual/capacity x100; dew point = saturation temp
Rainfall	Condensation + precipitation of moisture	3 types: Convectional, Orographic (relief), Cyclonic (frontal)

■ STATIC THEORY

• **Pressure belts (4 primary):** Equatorial Low (doldrums, 0-5 deg, thermal, calm ascending air) -> Sub-tropical High (25-35 deg, dynamic, horse latitudes, descending dry air -> deserts) -> Sub-polar Low (60-65 deg, dynamic) -> Polar High (thermal, cold dense air).

• **Planetary winds (Coriolis-deflected):** Trade Winds blow from Sub-tropical High to Equatorial Low (NE in N-hemisphere, SE in S). Westerlies from Sub-tropical High to Sub-polar Low (SW in N). Polar Easterlies from Polar High to Sub-polar Low.

• **Three types of rainfall (GC Leong):** (1) Convectional - heated air rises, cools, condenses (equatorial afternoons, thunderstorms). (2) Orographic/Relief - moist air forced up a mountain, windward wet + leeward rain-shadow dry. (3) Cyclonic/Frontal - warm & cold air masses meet, warm air lifted along the front (temperate depressions, tropical cyclones).

• **Humidity concepts:** Absolute humidity = actual mass of vapour/volume. Relative humidity = ratio of actual to maximum the air can hold at that temperature. Warm air holds more moisture. When RH reaches 100% = dew point; further cooling -> condensation (dew, fog, cloud).

• **Condensation forms:** Dew, frost, fog, mist, clouds. Clouds classified by height (high: cirrus; middle: alto; low: strato; vertical: cumulonimbus = thunderstorm/rain cloud).

■ MUST-KNOW FACTS

■■ UPSC TRAPS

1. Doldrums = equatorial low pressure belt, calm ascending air, region of convectional rainfall. **2.** Horse Latitudes = sub-tropical high pressure (30 deg), descending dry air -> world's great deserts (Sahara, Thar, Kalahari). **3.** Coriolis force is zero at equator, maximum at poles - deflects winds right in N-hemisphere, left in S-hemisphere (Ferrel's Law). **4.** Windward slope = heavy orographic rain; leeward slope = rain-shadow (e.g. Western Ghats windward wet, Deccan leeward dry). **5.** Cumulonimbus = towering vertical cloud associated with thunderstorms, hail and heavy rain.	✗ Pressure belts are permanent and fixed in latitude. ✓ They shift ~5 deg north/south seasonally following the apparent movement of the sun (northward in June, southward in December) - this shift drives monsoons. ✗ Trade winds and Westerlies blow towards the equator alike. ✓ Trades blow towards the equator (Sub-tropical High -> Equatorial Low); Westerlies blow away from equator (Sub-tropical High -> Sub-polar Low). ✗ High relative humidity always means high absolute moisture. ✓ RH is temperature-dependent; cold air with little vapour can still be at 100% RH. Absolute humidity can be low even at 100% RH.

■ **MAINS ANGLE**

Explain how the differential heating of the earth's surface sets up the global pressure-and-wind system, and how the seasonal migration of pressure belts governs the Indian monsoon.

■ **STUDY LINK**

Geography -> Climatology -> Atmospheric Circulation (feeds directly into Ch-14 Climate Classification and the Monsoon mechanism)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: India's 2026 Heatwave - Heat Dome, Loo Winds & Anticyclonic Subsidence

GS-I (Climatology) + GS-III (Disaster Management)
Geography - Current Affairs

■ **NEWS TRIGGER**

May-June 2026: India recorded extreme heat (Sri Ganganagar, Rajasthan touched ~48.2 deg C; IMD flagged 97 of the world's 100 hottest cities in India). A persistent 'heat dome' high-pressure system trapped hot air over the Indo-Gangetic plains; peak power demand hit a record 270.82 GW.

■ **INTRO & ORIGIN**

The 2026 heatwave is a textbook real-world case of the static pressure-and-wind concepts: an anticyclone (high pressure) causes subsidence (descending, compressed, warming air), suppresses cloud formation, weakens winds and blocks cooling western disturbances - a 'heat dome'.

IMD heatwave criteria (plains): max temp >=40 deg C AND departure of 4.5-6.4 deg C above normal, OR absolute max >=45 deg C. Severe heat wave: departure >6.4 deg C or actual max >=47 deg C.

■ **KEY DATA**

Driver	Mechanism (links to static concept)	Impact
Heat Dome	Persistent anticyclone -> subsidence -> adiabatic warming, no clouds	Trapped hot air over Indo-Gangetic plains
Loo winds	Hot dry NW winds, steep pressure gradient in May-June	Dehydration, heatstroke, high mortality

Driver	Mechanism (links to static concept)	Impact
Weak Western Disturbances	Fewer moisture-bearing systems -> no relief rain	Prolonged dry heat spell
Urban Heat Island	Concrete + low green cover retains heat	Cities hotter than surrounding rural areas

■ STATIC THEORY

• **Loo** = hot, dry, gusty afternoon wind blowing over NW India (Rajasthan, Punjab, Haryana, UP) in May-June; caused by a steep pressure gradient (~1.0-1.5 mb per degree latitude - D.R. Khullar) as NW India heats intensely.

• **Anticyclone / Heat Dome:** a high-pressure system with descending and diverging air; subsidence compresses and warms air adiabatically, prevents convection and cloud formation - hence clear skies and intense surface heating.

• **IMD colour-coded alerts:** Green (no action) -> Yellow (watch) -> Orange (be prepared) -> Red (take action).

■ MUST-KNOW FACTS

1. IMD plains heatwave threshold: max >=40 deg C with departure 4.5-6.4 deg C, or absolute >=45 deg C; severe if >=47 deg C.

2. Loo is a LOCAL wind (thermally induced), not a planetary wind - blows strongest 3-6 pm, May-June.

3. Heat dome = anticyclonic subsidence -> exactly the sub-tropical high pressure mechanism that creates deserts.

4. Record peak power demand 2026: 270.82 GW (grid stress from cooling load).

5. Night-time minimum temperatures rising ~0.21 deg C/decade (IMD) - reduces overnight recovery, worsens health impact.

■■ UPSC TRAPS

✗ A heat dome is a type of cyclone.

✓ It is an anticyclone (high pressure) with descending air - the opposite of a cyclone's ascending low-pressure air.

✗ Loo is a monsoon wind.

✓ Loo is a pre-monsoon local wind of the hot dry season (May-June); it stops once the SW monsoon arrives.

■ MAINS ANGLE

The 2026 heat dome shows how a persistent anticyclone can convert a normal summer into a disaster - link atmospheric subsidence, urban heat islands and the need for Heat Action Plans (GS-III DM).

■ STUDY LINK

Geography -> Climatology -> Pressure & Winds (anticyclone/subsidence) + local winds (Loo); connect to GS-III NDMA Heat Action Plans

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End of Register Notes — UPSC Agent / Copilot CLI